Hoek on questions and decisions

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1. Goals and strategy

Ultimately, Dan is looking for a model of the mind that is less idealized than standard decision theory, and accordingly capable of fitting/predicting a wider range of actual human behavior.

This paper focuses on the case of "full belief"; the second half of Dan's dissertation handled the generalization to partial belief. Full belief turns out to be the special case of credence 1.

The basic strategy is to take the objects of belief (and credence) to be *quizpositions*—items modelled as ordered pairs <Q,A> (written A^{Q}) where Q is a partition of the set of possible worlds and $\mathsf{A} \subseteq \mathsf{Q}$. The idea is that by conceiving of the objects of belief in this way, we will be able to formulate principles about how beliefs constrain decision-making that don't require such a massive invocation of ceteris-paribus clauses.

2. The inquisitive belief-action principle

The inquisitive belief-action principle: A belief that A^{Q} manifests in action as a disposition to forego A^{Q}-dominated options in any decision situation that raises Q.

Definitions:

- An **option** is a real-valued function from possible worlds to real numbers.
- A **decision problem** is a finite set of options.
- Decision problem Δ **faithfully represents** an agent's choice iff there is a mapping between the elements of Δ and the agent's available actions such that for each world w and $\mathbf{a} \in \Delta$, $\mathbf{a}(w)$ = the utility of the world that, at w, would have obtained had the agent performed the action corresponding to $\mathbf{a}$.
- Decision problem Δ **raises** Q iff for every $\mathbf{a} \in \Delta$, $\mathsf{q} \in \mathsf{Q}$, and $w_1, w_2 \in \mathsf{q}$, $\mathbf{a}(w_1) = \mathbf{a}(w_2)$.
- When Δ is a decision problem and $\mathbf{a}, \mathbf{b} \in \Delta$, $\mathbf{a}$ strictly A^{Q}-dominates $\mathbf{b}$ iff $\mathbf{a}(w) > \mathbf{b}(w)$ whenever $w \in \mathsf{q}$ and $\mathsf{q} \in \mathsf{Q}$.

Key point: the action-disposition associated with a belief only kicks in when the agent is in a decision problem associated with the corresponding question. Even if the agent believes some quizposition intensionally equivalent to A^{Q}, they might still end up choosing an A^{Q}-dominated option.

How this applies to examples like *Romeo Recall* and *Trivial Trouble*.

3. A worry

The standard practice of characterizing real-world decisions using decision matrices with some small number of states already involves an implicit appeal to the agent's beliefs (or knowledge). If we are trying to characterize the decision problem faced by an agent in a way that's independent of the agent's beliefs, we'll need to include tons more states that the agent knows don't obtain—e.g. ones where the Sphinx asked a different riddle. This means that according to the above definitions, the question raised by any decision that a real agent faces will always be absolutely enormous and super-complex.

Do agents typically have beliefs in such big quizpositions?

If so, the agent's failure to form any beliefs that address the questions they face in examples like *Trivial Trouble* seems mysterious.

4. Coherent information states

A predictive theory will need to place some restrictions on the combinations of beliefs an agent can maintain together. But we want something less demanding than closure under entailment and conjunction.

Inquisitive Belief States: an agent's beliefs form a *coherent inquisitive information state*.

Defs:

- an *inquisitive information state* is a set of quizpositions $\mathbf{I}$ subject to the following closure conditions:

  (i) Closure under parthood: if $A^Q \in \mathbf{I}$ and A^Q contains B^R, then $B^R \in \mathbf{I}$.

  (ii) Partial closure under conjunction: if $A^Q \in \mathbf{I}$ and $B^R \in \mathbf{I}$ and Q contains R, then $AB^Q \in \mathbf{I}$.

  or equivalently:

  (iia) Very restricted closure under conjunction: if $A^Q \in \mathbf{I}$ and $B^Q \in \mathbf{I}$ then $AB^Q \in \mathbf{I}$.

  (iib) Upward closure: If $B^R \in \mathbf{I}$ and $\mathbf{I}$ contains any quizpositions about Q, $B^Q \in \mathbf{I}$.

- The information state is *coherent* iff it contains no quizposition of the form $\varnothing^Q$.
- Q contains R (is coarser than) iff every R-cell is a union of Q-cells.
- A^Q contains B^R iff Q contains R and $\cup A \subseteq \cup B$.
- $AB^{QR} = \langle QR, AB \rangle = \langle \{(q \cap r : q \in Q, r \in R\} \setminus \{\varnothing\}, \{(a \cap b : a \in A, b \in B\} \setminus \{\varnothing\} \rangle$

Note: there will be an analogous quizpositional weakening of the standard notion of *probabilistic coherence*. It requires that (i) for every question, if one has any credences in quizpositions about that question, then one's credences about them all, and these obey the probability axioms, and (ii) whenever one has credences about a question, one has the same credences about any coarser question.

5. How we can have inconsistency without incoherence

A coherent agent can have two inconsistent beliefs, so long as (i) they are about different questions; (ii) they don't have any beliefs about any fine-grained question that contains both questions, and (iii) for any coarser question contained in both questions, the results of projecting the two quizpositions down to that coarser question are consistent.

In such a case, there is no coherent information state that includes any beliefs about the finer-grained question. Picture: if the agent even came to believe the necessary truth Q^{Q}, they would have to undergo a complex process of belief-revision.

Even for a non-inconsistent belief state, expanding to include Q^{Q} can rule out possible worlds that weren't ruled out before. This is supposed to model of what's going on in deduction.

6. The Duck Principle

The claim that no agent believes any quizposition of the form $\varnothing^{\mathsf{Q}}$ can be motivated directly from the Inquisitive Belief-Action Principle. *Every* option in a decision problem that raises Q is, vacuously, $\varnothing^{\mathsf{Q}}$-dominated. So an agent who believed $\varnothing^{\mathsf{Q}}$ would have be disposed, when facing such a decision problem, to forego all of its options. But such a disposition is impossible.

The other clauses of Inquisitive Belief Statescan be motivated from the Inquisitive Belief-Action Principle together with a kind of highly-qualified sort of behaviorism:

> *Duck Principle:* If an agent X has the behavioral dispositions that are associated with a belief with a certain content, and moreover X has those dispositions in virtue of their beliefs, then X actually does have a belief with that content.

7. Behavioral argument for Inquisitive Belief States

This is very straightforward for *closure under parthood*. Suppose that the agent believes A^{Q} and A^{Q} contains B^{R}. Since Q contains R, any decision situation that raises R raises Q; moreover, any B^{R}-dominated action is A^{Q}-dominated. So in any such decision situation the agent is disposed to forego all B^{R}-dominated options. So by the Duck Principle, the agent believes that B^{R}.

For *partial closure under conjunction*, it requires a Dutch Book argument (footnote 25). I'm going to paste it here since it seems really important.

[31] *Proof.* The argument is analogous to that in footnote 25. Suppose our agent X fails to avoid AB^{Q}-dominated options faced with Q. Then X will sometimes pick $\mathbf{b}$ over the alternative $\mathbf{a}$ in a Q-raising choice Δ where $\mathbf{b}(\mathsf{q}) < \mathbf{a}(\mathsf{q})$ for all $\mathsf{q} \in \mathsf{AB}$. On this occasion, we offer X a bet $\mathbf{t}$ that yields some small utility ϵ if B^{R} is true and great disutility $-C$ if B^{R} is false. Note R addresses $\{\mathbf{o}, \mathbf{t}\}$, and $\mathbf{t}$ B^{R}-dominates $\mathbf{o}$. But to **t**ake the bet would be to choose $\mathbf{b} + \mathbf{t}$ which is A^{Q}-dominated by $\mathbf{a} + \mathbf{o}$. Moreover, the composite decision problem consisting of Δ and $\{\mathbf{o}, \mathbf{t}\}$ is addressed by QR, and $\mathsf{QR} = \mathsf{Q}$ because R is part of Q. So if they **t**ake the bet, X chooses an A^{Q}-dominated option faced with Q, while to leave the bet is to choose a B^{R}-dominated option when faced with R. $\square$

8. A worry about partial closure under conjunction

In cases like *Romeo Recall*, we predict that an agent who believes *Romeo's number is NNN* as an answer to the polar question and has any beliefs at all about *What is Romeo's number?* will also believe *Romeo's number is NNN* as an answer to the big question.

This is surprising: it seems intuitively like an agent with the relevant limitations could still believe some things about the big question, e.g. *Romeo's number isn't 1-888-8888888*.

And there is also trouble coming from the Duck Principle. For couldn't an agent with the recall limitation be disposed never to choose any option dominated by *Romeo's number isn't 1-888-8888888* when faced with the big question? If they were, the Duck Principle says that they do have that belief.

It seems that partial closure under conjunction is blocking our ability to explain the real phenomenon where agents do better when they have fewer options because the computational challenge of searching through the options is more tractable.

9. So, what about that Dutch Book argument?

Couldn't it be true that the agent will sometimes choose pq-dominated options when faced with the big question Q, but is disposed to only do so in decision situations that don't also raise the smaller question R?

10. Miscellaneous other questions

- How can we model changing your beliefs in response to new information?
- Does this help with worries about the computational complexity of Bayesianism?